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H5 — Combinatorial Game Theory

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Chapter 1

Introduction to Combinatorial Game Theory

1.1 Overview

Combinatorial Game Theory (CGT) is the mathematical study of deterministic two-player games with perfect information and no element of chance. In such games, both players have complete knowledge of the current game state, and the outcome depends entirely on the decisions made by the players.

Unlike classical game theory, which is often concerned with economics, auctions, and strategic interactions involving uncertainty, combinatorial game theory focuses on games such as Nim, Wythoff Nim, Domineering, Hackenbush, and many endgames of Chess and Go.

The primary objective of combinatorial game theory is to determine optimal play. Given a game position, we seek to answer questions such as:

- Which player wins under perfect play?
- What move should the current player choose?
- Can a complicated game be reduced to a simpler mathematical object?

The answers to these questions often reveal surprisingly deep mathematical structures. Throughout the development of CGT, researchers have discovered connections with algebra, graph theory, number theory, logic, and computer science. The subject demonstrates one of the central themes of mathematics: simple rules can generate remarkably sophisticated structures.

1.2 Characteristics of Combinatorial Games

A game is generally considered a combinatorial game if it satisfies the following properties:

- **Two Players** — exactly two players participate, traditionally called Left and Right.
- **Perfect Information** — both players know the complete state of the game at all times; there is no hidden information.
- **No Randomness** — the game contains no dice rolls, shuffled cards, or random events.
- **Alternating Moves** — players take turns making moves.
- **Finite Termination** — every sequence of legal moves eventually ends.

- **Deterministic Outcome** — the outcome depends solely on the choices made by the players.

Many well-known games satisfy some of these properties but not all of them. Poker, for example, involves hidden information and randomness and therefore lies outside the scope of classical combinatorial game theory.

1.3 Examples of Combinatorial Games

Several games have played important roles in the development of CGT:

- **Nim** — the most fundamental impartial game and the foundation of modern combinatorial game theory.
- **Wythoff Nim** — an extension of Nim that introduces diagonal moves and leads to surprising connections with the Golden Ratio.
- **Hackenbush** — a graph-based game that bridges impartial games, partizan games, and Conway's theory of numbers.
- **Domineering** — a partizan game in which players place dominoes in different orientations.
- **Clobber** — a board game involving colored pieces and asymmetric move sets.

1.4 Historical Development

The origins of combinatorial game theory can be traced to the beginning of the twentieth century.

- **Charles Bouton and Nim** — in 1901, Bouton completely solved the game of Nim. His solution introduced the concept of the nim-sum and remains one of the most elegant results in recreational mathematics.
- **Sprague and Grundy** — during the 1930s, Roland Sprague and Patrick Grundy independently discovered that every impartial game behaves like a Nim heap of some size, transforming Nim into a universal model for all impartial games.
- **John Conway** — the modern era of CGT began with Conway's work introducing numbers as games, canonical forms, game comparison, and surreal numbers.
- **Winning Ways** — the publication of *Winning Ways for Your Mathematical Plays* by Berlekamp, Conway, and Guy established CGT as a major mathematical discipline.

1.5 Why Study Combinatorial Game Theory?

At first glance, combinatorial games may appear to be recreational puzzles. However, the mathematical ideas that emerge from them have broad significance:

- **Mathematical Beauty** — Wythoff Nim connects to the Golden Ratio; Conway's theory leads to surreal numbers.
- **Algorithm Design** — game-solving techniques often inspire efficient algorithms.

- **Artificial Intelligence** — modern game-playing systems rely on game trees, evaluation functions, and optimal decision making.
- **Graph Theory** — several combinatorial games are naturally represented using graphs.
- **Theoretical Computer Science** — the computational complexity of games is an active research area.

1.6 Structure of This Report

This report presents a Summer of Science project on Combinatorial Game Theory. It begins with winning and losing positions and develops the theory of impartial games through Nim, Wythoff Nim, and Sprague–Grundy Theory. It then introduces Hackenbush and Conway's theory of games, including numbers as games, canonical forms, and game comparison, before turning to partizan games. The report continues with graph games, examining Geography, Node Kayles, and Treblecross, and returns to partizan territory through Toads and Frogs and Blue–Red Hackenbush, which motivate infinitesimal game values. It closes with temperature theory and a computational implementation of these ideas in C++.

The objective is not only to present results but also to develop the mathematical intuition required for more advanced topics in modern combinatorial game theory.

1.7 Conclusion

Combinatorial Game Theory occupies a unique position at the intersection of mathematics, computer science, and logic. Beginning with simple games such as Nim, the subject develops into a rich mathematical framework capable of describing numbers, strategies, and complex decision processes. The chapters that follow build this theory systematically, beginning with the classification of winning and losing positions and progressing toward Conway's remarkable theory of games and numbers.

Chapter 2

P-Positions and N-Positions

2.1 Introduction

Before attempting to solve any combinatorial game, it is necessary to understand what it means for a position to be winning or losing. The concepts of P-positions and N-positions provide a universal framework for analyzing game positions and form the foundation upon which much of combinatorial game theory is built. Almost every major result in CGT — Bouton's theorem, Sprague–Grundy theory, and Conway's theory of games — relies on this classification.

The central question is simple: given a game position, can the player whose turn it is force a win? To answer this systematically, we introduce P-positions and N-positions.

2.2 Winning and Losing Positions

Consider a game played under the normal-play convention, meaning the player making the final legal move wins. Every position falls into one of two categories.

- **P-Position (Previous Player Winning)** — a position from which the player whose turn it is cannot force a win; commonly called a losing position.
- **N-Position (Next Player Winning)** — a position from which the player whose turn it is can force a win; commonly called a winning position.

2.3 Recursive Characterization

The power of P-positions and N-positions comes from their recursive definition.

1. Rule 1 — every terminal position (no legal moves possible) is a P-position.
2. Rule 2 — a position is an N-position if at least one legal move leads to a P-position.
3. Rule 3 — a position is a P-position if every legal move leads to an N-position.

These three rules completely characterize winning and losing positions.

2.4 Example: A Simple Stone Removal Game

Consider a game with a pile of stones in which a player may remove one or two stones; the player making the final move wins.

- Position 0 — no move possible $\Rightarrow$ P-position.

- Position 1 — moves to 0 (a P-position) ⇒ N-position.
- Position 2 — moves to 1 or 0; 0 is a P-position ⇒ N-position.
- Position 3 — moves to 2 or 1, both N-positions ⇒ P-position.
- Position 4 — moves to 3 (a P-position) or 2 ⇒ N-position.

The repeating pattern becomes: P, N, N, P, N, N, P, N, N, . . . — a clean illustration of how recursive reasoning completely solves a game.

2.5 Game Trees

The recursive structure of games is often represented as a game tree. For the subtraction game above, position 4 branches to 3 (a P-position) and 2, which itself branches to 1 and 0. Labels are assigned from the bottom upward — a process known as backward induction.

2.6 Backward Induction

Backward induction proceeds in a few simple steps:

4. Identify terminal positions.
5. Label them as P.
6. Move one step upward.
7. Apply the recursive rules.
8. Continue until the initial position is classified.

This technique reappears throughout combinatorial game theory.

2.7 Importance of P-Positions

- They identify losing states — once a player reaches a P-position, optimal play becomes easier to analyze.
- They determine winning strategies — a winning strategy typically consists of moving from an N-position to a P-position.
- They generalize across games — the same ideas recur in Nim, Wythoff Nim, Sprague–Grundy theory, graph games, and Conway games.

2.8 Strategic Interpretation

Winning positions are positions from which a move to a losing position exists; losing positions are positions from which every move leads to a winning position. This simple observation drives much of modern combinatorial game theory.

2.9 Connection to Nim

The next chapter studies Nim, the most important impartial game. One of Bouton's central achievements was a complete characterization of all P-positions in Nim — rather than labeling positions one at a time, he found a direct criterion involving binary arithmetic and the XOR operation, allowing positions with millions of tokens to be analyzed instantly.

2.10 Summary

This chapter introduced P-positions and N-positions and their recursive characterization, demonstrated through a simple subtraction game solved by backward induction. These ideas form the conceptual foundation for the analysis of Nim and every major theory that follows.

Chapter 3

The Game of Nim

3.1 Introduction

Among all combinatorial games, Nim occupies a unique position: it is the simplest non-trivial impartial game and serves as the foundation upon which much of modern combinatorial game theory is built. Sprague–Grundy theory later demonstrates that every impartial game can be represented as a Nim heap of appropriate size, so understanding Nim is essential to understanding the broader framework of CGT. The game was completely solved by Charles L. Bouton in 1901, in one of the most elegant applications of binary arithmetic in mathematics.

3.2 Rules of Nim

A Nim position consists of several heaps of tokens. Two players alternate turns; during a move, a player chooses exactly one heap and removes any positive number of tokens from it. The player making the final move wins.

Example — consider three heaps $(3, 4, 5)$. Possible moves include $(2, 4, 5)$, $(1, 4, 5)$, $(0, 4, 5)$, $(3, 2, 5)$, $(3, 4, 1)$, and many others; the challenge is determining whether the position is winning or losing.

3.3 Small Examples

- Single heap — (0) has no legal move, so it is a P-position; (1) lets the player take the token and win, so it is an N-position.
- Two equal heaps — $(1, 1)$ can only move to $(0, 1)$ or $(1, 0)$, both N-positions, so $(1, 1)$ is a P-position. Likewise $(2, 2)$ and $(3, 3)$ are P-positions, hinting at a deeper structure.

3.4 Binary Representation

Bouton's key insight was that ordinary arithmetic is not the right tool for analyzing Nim — heap sizes should instead be represented in binary. For example, $3 = 011$, $4 = 100$, and $5 = 101$ in binary; the relevant operation is XOR rather than addition.

3.5 The Nim-Sum

The nim-sum of heap sizes is defined as their bitwise XOR.

$$011 \text{ XOR } 100 \text{ XOR } 101 = 010 \quad \Rightarrow \quad 3 \oplus 4 \oplus 5 = 2$$

The nim-sum completely determines the nature of a Nim position.

3.6 Bouton's Theorem

$$\text{A Nim position is a P-position} \iff a_1 \oplus a_2 \oplus \cdots \oplus a_n = 0$$

Proof — the argument has two parts.

- **Part I:** every move from a zero nim-sum position produces a non-zero nim-sum position. If $a_1 \oplus \cdots \oplus a_n = 0$ and a legal move changes exactly one heap a_k, the XOR value must change, so the new nim-sum cannot remain zero.
- **Part II:** every non-zero nim-sum position has a move to a zero nim-sum position. Let $S = a_1 \oplus \cdots \oplus a_n \neq 0$, and let the most significant set bit of S be in some heap a_i. Setting $a_i' = a_i \oplus S$ removes that bit, so $a_i' < a_i$ and the move is legal. The new nim-sum is $S \oplus a_i \oplus (a_i \oplus S) = 0$, using $S \oplus S = 0$ and $a_i \oplus a_i = 0$.

Combining Parts I and II proves the theorem: the nim-sum is zero if and only if the position is a P-position.

3.7 Worked Examples

- $(1, 4, 5)$: $001 \oplus 100 \oplus 101 = 000 \Rightarrow$ P-position; the second player wins under optimal play.
- $(3, 4, 5)$: $011 \oplus 100 \oplus 101 = 010$ $(= 2) \Rightarrow$ N-position. Taking heap 3 to $3 \oplus 2 = 1$ gives $(1, 4, 5)$, whose nim-sum is zero — the correct winning move.
- $(7, 10, 13)$: $0111 \oplus 1010 \oplus 1101 = 0000 \Rightarrow$ P-position.

3.8 General Winning Strategy

- If nim-sum $= 0$ — you are in a losing position; no winning move exists against perfect play.
- If nim-sum $\neq 0$ — reduce an appropriate heap so the resulting nim-sum becomes zero, and repeat after every opponent move until the opponent is forced into the terminal position.

3.9 Importance of Nim

- Foundation of Sprague–Grundy theory — every impartial game can be reduced to Nim.
- Introduction of XOR arithmetic — a powerful application of binary operations.
- Connection to algorithms — many game-solving algorithms are Nim-like in structure.
- Green Hackenbush — Green Hackenbush positions correspond directly to Nim heaps.

3.10 Summary

This chapter developed the complete mathematical solution of Nim through the nim-sum and Bouton's theorem, which characterizes all winning and losing positions and provides an explicit winning strategy whenever the nim-sum is non-zero. The next chapter studies Wythoff Nim, revealing surprising connections between combinatorial games, irrational numbers, and the Golden Ratio.

Chapter 4

Sprague–Grundy Theory

4.1 Motivation

Nim possesses a complete mathematical solution through Bouton's theorem, but most impartial games do not resemble Nim — subtraction games, graph games, token-moving games, and countless recreational puzzles look nothing alike on the surface. A natural question arises: can every impartial game be reduced to Nim? Roland Sprague and Patrick Grundy showed the answer is yes — every finite impartial game behaves exactly like a Nim heap of some size, transforming Nim from a single game into the universal model for all impartial games.

4.2 The Mex Function

The fundamental building block of Sprague–Grundy theory is the minimum excluded value function, or mex. For a set of non-negative integers S, $\mathrm{mex}(S)$ is the smallest non-negative integer not in S.

$$\mathrm{mex}(S) = \min\{\, n \geq 0 : n \notin S \,\}$$

- $\mathrm{mex}(\{1, 2, 3\}) = 0$, since 0 is missing.
- $\mathrm{mex}(\{0, 1, 3\}) = 2$, since 0 and 1 are present but 2 is absent.
- $\mathrm{mex}(\{0, 1, 2, 3\}) = 4$.
- $\mathrm{mex}(\emptyset) = 0$.

4.3 Grundy Numbers

Every impartial game position P is assigned a non-negative integer, its Grundy number, defined recursively over the positions Q reachable in one move:

$$g(P) = \mathrm{mex}\{\, g(Q) : Q \text{ reachable from } P \,\}$$

To compute a Grundy value: (1) compute the Grundy values of all reachable options, (2) collect them into a set, and (3) apply mex.

4.4 Terminal Positions

A terminal position P has no legal moves, so $g(P) = \mathrm{mex}(\emptyset) = 0$ — every terminal position has Grundy value zero, mirroring an empty Nim heap.

4.5 Example: Remove 1 or 2 Stones

For a pile where a player may remove one or two stones, let $g(n)$ be the Grundy number of a pile of n stones:

$$\begin{aligned} g(0) &= 0 \\ g(1) &= \text{mex}\{g(0)\} = \text{mex}\{0\} = 1 \\ g(2) &= \text{mex}\{g(1), g(0)\} = \text{mex}\{1, 0\} = 2 \\ g(3) &= \text{mex}\{g(2), g(1)\} = \text{mex}\{2, 1\} = 0 \\ g(4) &= \text{mex}\{g(3), g(2)\} = \text{mex}\{0, 2\} = 1 \\ g(5) &= \text{mex}\{g(4), g(3)\} = \text{mex}\{1, 0\} = 2 \end{aligned}$$

The resulting sequence is $0, 1, 2, 0, 1, 2, 0, 1, 2, \ldots$ — perfectly periodic with period 3.

4.6 The Sprague–Grundy Theorem

$$P \cong *g(P) \qquad \text{— every impartial game equals a Nim heap of size } g(P)$$

This result is often called the Fundamental Theorem of Impartial Games.

4.7 Interpretation

If a complex position P evaluates to $g(P) = 5$, the entire position behaves exactly like a single Nim heap of size 5. $g(P) = 0$ means the position behaves like an empty Nim heap — a losing P-position. Every impartial game can therefore be conceptually replaced by an equivalent Nim heap.

4.8 Sums of Games

If a game is composed of independent components with Grundy values $g_1, \ldots, g_n$, the combined position's total Grundy value is their nim-sum:

$$g = g_1 \oplus g_2 \oplus \cdots \oplus g_n$$

This is one of the most powerful and useful consequences of Sprague–Grundy theory.

4.9 Significance of the Theorem

Before Sprague–Grundy theory, every unique impartial game required separate, custom mathematical analysis. After it, every impartial game effectively becomes an equivalent game of Nim, so the entire study of impartial games reduces to two uniform operations:

9. Computing localized Grundy numbers recursively.
10. Applying standard binary XOR arithmetic across concurrent games.

This universal reduction stands as one of the most elegant concepts in game theory.

4.10 Summary

Sprague–Grundy theory demonstrates that every impartial game behaves exactly like a Nim heap. Grundy numbers give a complete description of impartial positions, and XOR arithmetic determines the winner of combined games. The next chapter examines applications, including Kayles and Dawson's Kayles.

Chapter 5

Applications of Sprague–Grundy Theory

5.1 Introduction

The Sprague–Grundy theorem is powerful because it converts a wide variety of games into Nim. Once the Grundy number of every position is known, the entire game can be analyzed through XOR arithmetic, and many apparently unrelated games turn out to be mathematically identical to Nim.

5.2 Subtraction Games

Suppose a player may remove a number of stones from a finite set $S = \{1, 3, 4\}$ from a single pile, with $g(n)$ the Grundy value of a pile of size n:

$$g(n) = \text{mex}\{\, g(n-1), g(n-3), g(n-4) \,\}$$

The first values are:

n	0	1	2	3	4	5	6	7
$g(n)$	0	1	0	1	2	3	2	0

such sequences often become periodic for larger n, a phenomenon studied extensively in search of analytical shortcuts.

5.3 DAG Games

Consider a directed acyclic graph with a token on a starting vertex; a legal move slides the token along a directed edge. The player unable to move loses. Terminal vertices (out-degree zero) have $g(v) = 0$; every other vertex takes $g(v) = \text{mex}$ of its children's Grundy values. This framework appears frequently in competitive programming, where many problems reduce directly to computing Grundy numbers over explicit graphs.

5.4 The Game of Kayles

Kayles — one of the first non-trivial games successfully analyzed with Grundy numbers — is played on a row of bowling pins. A move removes either one pin or two adjacent pins.

Example: a row of seven pins, |||||||, becomes ||| ||| after the middle pin is taken. The single row splits into two independent games — exactly the disjoint situation Sprague–Grundy theory was designed for.

5.5 Game Analysis of Kayles

If a move splits a row into segments with Grundy values g_1 and g_2, the resulting position's value is $g_1 \oplus g_2$. The total Grundy value of the original row is the mex over all such resulting XOR values:

$$g(n) = \text{mex}(\text{ all resulting XOR values })$$

This recursive procedure completely solves Kayles for any row length.

5.6 Dawson's Kayles

Dawson's Kayles alters the removal rule: choosing a pin destroys it along with its immediate neighbors. For seven pins, choosing the center pin removes three pins at once, producing $||\ \ ||$. As with ordinary Kayles, the row splits into independent components, so Sprague–Grundy theory applies naturally.

5.7 Importance of Dawson's Kayles

Dawson's Kayles is notable because its Grundy sequence exhibits long, non-trivial periodic structure. It remains an important testing ground for Sprague–Grundy methods and computational game theory.

5.8 General Strategy for Impartial Games

11. Define the distinct game states and legal moves.
12. Compute Grundy values recursively from the terminal positions upward.
13. Combine independent components using the bitwise XOR operation.
14. Determine whether the final nim-sum is zero (P-position) or non-zero (N-position).

5.9 Summary

Sprague–Grundy theory transforms complicated games into equivalent games of Nim. Kayles, Dawson's Kayles, subtraction games, and DAG games are all fully solved through Grundy numbers and XOR arithmetic. The next chapter develops the mathematical objects underlying these values: nimbers.

Chapter 6

Nimber Theory

6.1 Motivation

Throughout the previous chapters, Grundy numbers repeatedly behaved exactly like single Nim heaps, suggesting they are not merely convenient bookkeeping labels but mathematical objects in their own right. These algebraic objects are called nimbers.

6.2 Definition of a Nimber

A Nim heap containing n stones is denoted $*n$ and called a nimber — for example $*0, *1, *2, *3, *4$, and so on. By the Sprague–Grundy theorem, every finite impartial game position is equivalent to one of these fundamental nimbers.

6.3 Nimbers as Universal Values

If an abstract position P evaluates to $g(P) = 7$, then $P = *7$, meaning the game behaves exactly like a Nim heap of size seven regardless of its original rules — which makes nimbers the universal values of impartial games.

6.4 Nimber Addition

Nimber addition, representing games played in parallel, corresponds exactly to the bitwise XOR operation:

$$*a + *b = *(a \oplus b)$$

- $*4 + *7 = *3$ (since $4 \oplus 7 = 3$)
- $*5 + *5 = *0$ (since $5 \oplus 5 = 0$)

This elegant property explains why two identical game components placed side by side always cancel to a net zero value — a losing position for the next player.

6.5 Importance of Nimbers

- A complete mathematical classification of impartial positions.
- A formal algebraic explanation for why XOR arithmetic solves games.
- A bridge connecting combinatorial game theory with abstract algebra.

Chapter 7

Wythoff Nim and Variants of Nim

7.1 Beyond Ordinary Nim

Ordinary Nim is completely solved by XOR arithmetic. A natural question is what happens when the core rules are modified slightly — even small changes can disrupt the XOR structure and open entirely new fields of mathematics. The most famous example is Wythoff Nim.

7.2 Wythoff Nim

The game is played with two heaps of tokens. A player may remove any number of tokens from the first heap, any number from the second heap, or the same number from both heaps simultaneously. This third option — the diagonal move — distinguishes Wythoff Nim from ordinary Nim and introduces beautiful geometric properties.

7.3 Losing Positions

The first few P-positions of Wythoff Nim are $(0,0), (1,2), (3,5), (4,7), (6,10), (8,13), \ldots$. Unlike ordinary Nim, these safe positions cannot be characterized using simple XOR sums.

7.4 The Golden Ratio

The losing positions are instead generated by the Golden Ratio, $\varphi = (1+\sqrt{5})/2 \approx 1.618$. The n-th losing pair (a_n, b_n) is given directly by the floor function:

$$a_n = \lfloor n\varphi \rfloor \qquad b_n = \lfloor n\varphi^2 \rfloor$$

This remarkable intersection directly links combinatorial game theory with classical number theory and irrational quantities.

7.5 Beatty Sequences

The sequences $\lfloor n\varphi \rfloor$ and $\lfloor n\varphi^2 \rfloor$ form a pair of complementary Beatty sequences. By Beatty's theorem, these two sequences partition the positive integers perfectly — every positive integer appears exactly once among the coordinates of the losing pairs — ensuring a player can always force the game toward a losing configuration for the opponent.

7.6 Misère Nim

Misère Nim alters only the win condition: the player who takes the last token loses. Although this seems a minor change, the optimal strategy shifts dramatically near the end of the game. Bouton's theorem remains correct for almost the entire game, until the endgame, where a player must pivot to leave an odd number of size-one heaps.

> *Strategy: play normally while at least two heaps have size greater than one. Once fewer than two such heaps remain, play so that after your move an odd number of size-one heaps remain. This strategy forces a win in Misère Nim.*

7.7 Moore's Nim

Moore's Nim generalizes Nim by allowing a player to remove tokens from up to k independent heaps simultaneously. For $k = 1$ the game reduces to ordinary Nim; for larger k, analysis requires working in base $(k + 1)$ rather than binary — ordinary Nim is the special case $k = 1$, where base $(k + 1) = 2$.

7.8 Fibonacci Nim

Fibonacci Nim imposes a history-dependent restriction: on the first move a player cannot take the entire pile, and on subsequent turns a player may remove at most twice the number of stones the previous player removed.

The solution is deeply connected to Zeckendorf's theorem, which states that every positive integer can be uniquely represented as a sum of non-consecutive Fibonacci numbers. Losing heap sizes fall on the Fibonacci sequence $\{2, 3, 5, 8, 13, 21, 34, \ldots\}$.

7.9 Summary of Nim Variants

- Ordinary Nim leads directly to binary XOR arithmetic.
- Wythoff Nim leads to the Golden Ratio and Beatty sequences.
- Fibonacci Nim leads to Zeckendorf representations and Fibonacci numbers.

Small changes in game rules can lead to entirely new fields of mathematics — simple games provide an accessible gateway to deep, interconnected structures.

Chapter 8

Hackenbush

8.1 Introduction

The games studied so far have primarily involved piles of stones or abstract positions. While mathematically rich, they do not fully reveal the connection between combinatorial games and numbers. That connection emerges through Hackenbush, which bridges Nim, Sprague–Grundy theory, partizan games, Conway numbers, and surreal numbers — making it one of the most important games in modern CGT.

8.2 Rules of Hackenbush

The game is played on a graph of colored edges attached to a horizontal line called the ground. Players alternately remove edges; whenever an edge is removed, every component that becomes disconnected from the ground also disappears — a cascading effect that produces highly non-trivial behavior.

8.3 Green Hackenbush

In Green Hackenbush, all edges are green and either player may remove any edge, making it an impartial game. For a vertical bamboo stalk of length three, removing the bottom edge destroys the whole stalk, the middle edge leaves a stalk of length one, and the top edge leaves a stalk of length two — exactly the choices available from a Nim heap of size three.

8.4 Green Hackenbush and Nim

A bamboo stalk of length $n \cong$ a Nim heap of size n

Proof — removing the k-th edge from the bottom leaves a stalk of length $k-1$, so the options from a stalk of length n are $\{0, 1, \ldots, n-1\}$, giving $g(n) = \text{mex}\{0, 1, \ldots, n-1\} = n$.

8.5 Forests of Stalks

A forest of independent bamboo stalks combines its components' Grundy values through XOR. A forest with stalk lengths 2, 3, and 5 has combined value $g = 2 \oplus 3 \oplus 5 = 4$ — the entire forest behaves like a single Nim heap of size four, giving a beautiful graphical interpretation of Sprague–Grundy theory.

8.6 Blue-Red Hackenbush

In Blue-Red Hackenbush, Left may remove only blue edges and Right only red edges. The game is no longer impartial — it becomes partizan, and since the available options depend on whose turn it is, Sprague–Grundy theory no longer applies directly, forcing the development of new mathematical tools.

8.7 The Birth of Numbers

A single blue edge attached to the ground lets Left move to an empty game while Right cannot move at all; this position is denoted $1 = \{0 \mid\}$. A single red edge, symmetrically, gives $-1 = \{\mid 0\}$. Positive and negative integers thus emerge naturally from simple partizan positions.

8.8 Fractions from Games

More elaborate Hackenbush structures generate fractions — combinations evaluating exactly to 1/2, 1/4, 3/4, and infinitely many others. The appearance of fractional values inside a strategic game was one of Conway's most remarkable discoveries.

8.9 Significance

Hackenbush is the core bridge between impartial game theory and Conway's broader theory of games — through it we discover numbers as games, fractions as games, surreal numbers, and general partizan values, all developed systematically in the next chapter.

Chapter 9

Conway Theory and Numbers as Games

9.1 Motivation

Blue-Red Hackenbush reveals that certain game positions behave exactly like numbers, raising a natural question: can numbers themselves be defined using games? John Conway answered affirmatively, developing a theory in which numbers arise naturally from basic game rules.

9.2 Recursive Definition of a Game

$$G = \{\, L \mid R \,\}$$

where L is the set of Left options and R the set of Right options, each option itself a game — games are defined recursively from the ground up.

9.3 The Simplest Game

The simplest possible game, in which neither player has a legal move, is denoted $0 = \{|\}$. Since neither player can move, the player to move immediately loses under normal play, so this zero game represents perfect strategic balance.

9.4 Positive Numbers

A game where Left can move to zero and Right has no options, $1 = \{0 \mid\}$, strictly favors Left and represents the value 1. Inductively, $2 = \{1 \mid\}$ and $3 = \{2 \mid\}$, and all positive integers emerge recursively.

9.5 Negative Numbers

Symmetrically, games favoring Right define the negative integers: $-1 = \{\mid 0\}$, $-2 = \{\mid -1\}$, $-3 = \{\mid -2\}$, providing Right with successive moves while Left is left stranded.

9.6 Fractions

Conway showed that dyadic fractions arise when games are placed between two known values — a game where Left can move to 0 and Right to 1 simplifies to the value exactly halfway

between them:

$$\tfrac{1}{2} = \{0 \mid 1\} \qquad \tfrac{1}{4} = \{0 \mid \tfrac{1}{2}\} \qquad \tfrac{3}{4} = \{\tfrac{1}{2} \mid 1\}$$

Standard arithmetic and numbers thus emerge directly from game-theoretic structures.

9.7 Surreal Numbers

Repeated, infinite application of Conway's recursive construction generates integers, rationals, infinite numbers, and infinitesimals. The resulting structure — the surreal numbers — forms the largest ordered field known in mathematics, unifying transfinite numbers and standard reals.

9.8 Importance

Conway's theory demonstrates that numbers are not external, abstract objects imposed upon games; instead they emerge naturally from the structural foundations of games themselves — one of the most profound ideas in modern mathematics.

Chapter 10

Comparing Games and Fuzzy Positions

10.1 Comparing Games

Ordinary numbers satisfy strict order relations — less than, greater than, or equal. Games possess a much richer structure: because players can have different options, not all games can be cleanly compared on a single number line.

10.2 Positive Games

A game G is positive ($G > 0$) if Left can force a win playing first or second — examples include 1, 2, and $1/2$.

10.3 Negative Games

A game G is negative ($G < 0$) if Right can force a win playing first or second — examples include -1, -2, and $-1/2$.

10.4 Zero Games

A game equals zero ($G = 0$) if the second player to move can always force a win, regardless of who starts. The fundamental example is the empty game, $0 = \{|\}$.

10.5 The Game Star

The game where both players can move to 0 is called star, $* = \{0 \mid 0\}$. Whoever moves first moves to 0 and wins, since the opponent then has no move. The first player to move always wins in star — behavior fundamentally different from ordinary numbers.

10.6 Fuzzy Games

Because the first player always wins in star, it cannot be classified as positive, negative, or zero — it is fuzzy with zero, written $* \parallel 0$, meaning the two positions are fundamentally incomparable. This phenomenon has no analogue in ordinary numerical arithmetic.

10.7 Significance

The existence of fuzzy games shows that game values cannot be represented solely by traditional real numbers — a richer, multi-dimensional framework is required to map strategic structures, motivating the broader study of partizan game theory.

10.8 Summary

This chapter introduced Conway's recursive definition of games, numbers as games, surreal numbers, and fuzzy positions — moving past basic impartial games and establishing the conceptual foundations for modern partizan game analysis.

Chapter 11

Canonical Forms

11.1 Motivation

As Conway's theory developed, it became clear that many different game trees can represent exactly the same strategic situation. Consider $G = \{1, 0 \mid\}$ and $H = \{1 \mid\}$: although their trees differ, the option 0 in G is never strategically useful for Left, so both games represent the same value. This motivates simplified, minimal representations called canonical forms — playing a role similar to reduced fractions in ordinary arithmetic.

11.2 Dominated Options

If Left has two options A and B with $A \geq B$, Left will never choose B — it contributes nothing strategically and may be safely removed; such an option is dominated.

Example — in $G = \{1, 0 \mid\}$, since $1 > 0$ the move to 0 is strictly dominated, so $\{1, 0 \mid\}$ reduces to $\{1 \mid\}$, a strategically identical but structurally smaller form.

11.3 Reversible Moves

Redundancy also arises when a move can be immediately neutralized by the opponent. If Left moves from G to A, but Right can respond by moving back to G (or somewhere at least as favorable to Right), the move to A provides no lasting advantage — such moves are reversible and may be bypassed during canonical simplification.

11.4 Canonical Representation

A game is in canonical form when no dominated options and no reversible moves remain for either player. The result is as simple as possible while fully preserving the game's strategic value.

11.5 Importance

- Easier structural comparison between abstract games.
- Simpler, more streamlined inductive proofs.
- Efficient algebraic computations in software solvers.
- A clear, unified classification system for distinct game values.

Chapter 12

Sums of Games

12.1 Motivation

Many games naturally decompose into independent components — multiple Nim heaps, disconnected Hackenbush stalks, or isolated regions of a Domineering board. A player may move in one component without affecting the others, motivating the formal concept of a game sum.

12.2 Definition

For independent games G and H, the disjunctive sum $G + H$ allows a legal move to be a valid move in G or a valid move in H, leaving the other component unchanged.

12.3 Example

Consider $1+1$, where each component is $1 = \{0 \mid\}$. The first player moves one heap to 0, giving $1 + 0$; the second player then moves the remaining heap to 0, giving $0 + 0$, leaving the first player without a move — the second player wins. Because the second player wins regardless of who starts, $1 + (-1)$ evaluates to a net zero value under game cancellation.

12.4 Nim and XOR

For impartial games, the disjunctive sum corresponds exactly to bitwise XOR: if components have Grundy values $g_1, \ldots, g_n$, the total value is $g = g_1 \oplus \cdots \oplus g_n$ — an observation at the absolute heart of Sprague–Grundy theory.

12.5 Outcome Classes

Let G and H be games, with four fundamental outcome classes: $G > 0$ (Left always wins), $G < 0$ (Right always wins), $G = 0$ (second player wins), and $G \parallel 0$ (first player wins, fuzzy with zero).

- $G > 0$ and $H > 0 \Rightarrow G + H > 0$ — Left wins each component independently.
- $G < 0$ and $H < 0 \Rightarrow G + H < 0$ — Right wins each component independently.
- $G = 0$ and $H = 0 \Rightarrow G + H = 0$ — the second player wins by mirroring.
- $G > 0$ and $H = 0 \Rightarrow G + H > 0$ — the zero game contributes no advantage.

- $G < 0$ and $H = 0 \Rightarrow G + H < 0$ — likewise determined by G.
- $G \parallel 0$ and $H = 0 \Rightarrow G + H \parallel 0$ — the zero game does not alter the outcome.
- $G > 0$ and $H < 0 \Rightarrow$ the outcome depends on the relative strengths of G and H; no universal rule exists.
- $G > 0$ and $H \parallel 0 \Rightarrow G + H$ is > 0 or $\parallel 0$, not uniquely determined.
- $G < 0$ and $H \parallel 0 \Rightarrow G + H$ is < 0 or $\parallel 0$, not uniquely determined.
- $G \parallel 0$ and $H \parallel 0 \Rightarrow G + H$ can be > 0, < 0, $= 0$, or $\parallel 0$ — two fuzzy games can interact in complex ways.

12.6 Importance

Decomposing complex games into small, independent components is one of the most powerful analytical ideas in combinatorial game theory — researchers can solve small localized pieces individually and combine their values using uniform arithmetic, rather than tackling a complex position directly.

Chapter 13

Introduction to Partizan Games

13.1 Impartial vs Partizan Games

The distinction between game types depends on move symmetry:

- **Impartial games** — both players have exactly the same legal moves from any position (Nim, Wythoff Nim, Green Hackenbush).
- **Partizan games** — move sets differ depending on whose turn it is (Chess, Go, Domineering, Clobber, Blue-Red Hackenbush).

Partizan games are generally harder to analyze because their values cannot be reduced to a single non-negative integer or Nim heap.

13.2 Domineering

Domineering is played on a rectangular grid; Left places vertical dominoes (2×1) and Right places horizontal dominoes (1×2), with no overlaps. The first player unable to place a domino loses. On a small 2×2 board, if Left places a vertical domino it immediately blocks Right, winning instantly — spatial geometry, not numerical heap size, determines the outcome.

13.3 Why Domineering Matters

- Game values and fractional values.
- Temperature theory and cooling operations.
- Canonical forms and tree reductions.
- Infinitesimal values.

Many of Conway's breakthrough examples originate from simple configurations on a Domineering board.

13.4 Clobber

Clobber is played on a board filled with black and white pieces in a checkerboard pattern; Left controls white, Right controls black. A legal move takes one's own piece, moves it orthogonally onto an adjacent opponent piece, and removes that piece from the board. The game ends under normal play when a player has no legal move.

13.5 Strategic Complexity

Although Clobber's movement rules are exceptionally simple, positions quickly split into independent components exhibiting a surprisingly rich mathematical structure, and remain an active subject of research into winning positions and algorithmic complexity.

13.6 Importance of Partizan Games

Partizan games motivated many of Conway's deepest breakthroughs — numbers as games, fuzzy values, canonical forms, infinitesimals, and game temperature — representing the natural continuation of the impartial theory developed earlier in this report.

Chapter 14

Looking Ahead: Graphs and Partizan Games

The foundations built so far — P-positions, Nim, Sprague–Grundy theory, nimbers, Wythoff Nim, Hackenbush, Conway's numbers, canonical forms, and partizan games — extend naturally in two directions, taken up in the chapters that follow.

- **Graph Games (Chapters 16–18)** — the token-and-heap model of Nim is replaced by a graph, revisiting Geography, Node Kayles, and Treblecross, and asking whether Sprague–Grundy theory survives the jump from piles to graphs.
- **Partizan Games, Continued (Chapters 19–20)** — Toads and Frogs and Blue–Red Hackenbush push further into positions where Left and Right no longer have symmetric options, motivating numerical game values, infinitesimals, and Tiny and Miny games.
- **Temperature (Chapter 21)** — a quantitative measure of how urgently a move should be played, together with the operations of cooling and heating.
- **Computational CGT (Chapter 22)** — C++ implementations of mex, Grundy numbers, and Nim, translating the theory developed so far into working code.

A central, recurring theme of this study is that exceptionally simple rules can generate profoundly deep mathematics. Nim leads naturally to binary XOR arithmetic; Sprague–Grundy theory shows that all finite impartial games behave identically to a single Nim heap; Wythoff Nim reveals unexpected connections with the Golden Ratio and Beatty sequences; Hackenbush provides a physical medium from which numbers, fractions, and surreal numbers emerge naturally; and Conway's recursive framework unifies all of these discoveries within a single, elegant algebraic system.

The chapters so far establish a rigorous foundation for the study of graph games, infinitesimals, game temperature, and partizan mechanics that follows — demonstrating how combinatorial games provide a brilliant, accessible meeting point for algebra, graph theory, number theory, mathematical logic, and theoretical computer science.

Chapter 15

From Nim to Graph Games

15.1 Moving to Graph Games

Every game analyzed so far — subtraction games, Nim, Kayles, Dawson's Kayles — has essentially been a variation of the same concept: a pile of tokens, from which some tokens are removed according to a rule. Now we move from the concept of Nim to something structurally different: a graph. Instead of a heap that shrinks, the board is a picture — dots connected by lines — and a token sits on a vertex, or edges get deleted one at a time.

The motivating question for the week was: does the entire Grundy-number machinery — mex, Sprague–Grundy, the XOR sum rule — still work once the board is a graph instead of a stack of tokens?

Short answer, arrived at only after a full chapter of work: yes, mostly, and with some genuinely surprising complications along the way.

15.2 Why Nim Alone Isn't Enough

Nim works so cleanly because a pile of five tokens carries no memory of how it became a pile of five. It is featureless — completely characterized by a single integer. That featurelessness is the entire reason that XOR-ing pile sizes together solves the whole game.

Most games do not hand us that kind of simplicity. Three concrete graph games break it in three different ways:

- **Geography** — the state is "where has the token already visited", a much richer object than a single number; it depends on the whole history of the walk, not just a count.
- **Node Kayles** — a single move can eliminate several vertices simultaneously, since picking a vertex removes it and everything adjacent to it. Piles never behave this way; removing tokens from one pile never affects another pile.
- **Treblecross** — a single move can literally cut the board into two disconnected leftover pieces, and, as this report eventually shows in detail, the resulting pieces are not even independent of how they were created.

So Nim tells us exactly what to do once a game's Grundy number is known. It says nothing about how to compute that number for something genuinely new. Closing that gap is what the remainder of this report is about.

15.3 Goals for the Chapters Ahead

- Getting comfortable with games such as graph games and Treblecross.

- Checking that Grundy theory — mex, Sprague–Grundy, the XOR rule — survives the jump from piles to graphs, and finding an explicit algorithm for computing it.

The remaining chapters extend this foundation in two directions. Chapter 19 explains, briefly, where the impartial tools stop working — the moment a game stops being "fair" between the two players — partizan games, briefly introduced in Chapter 13. There we discuss two more partizan games, Toads and Frogs and Blue–Red Hackenbush, which eventually lead to infinitesimals, temperature theory, and advanced CGT.

15.4 Structure of the Chapters Ahead

Chapters 16 and 17 introduce graph games and develop the ability to use Sprague–Grundy theory on them. Chapter 18 applies this to two different games — Node Kayles and Treblecross. Chapter 19 introduces Toads and Frogs, studying partizan games in more detail and eventually discussing numerical values of games. Chapter 20 discusses Blue–Red Hackenbush, and Chapter 21 leads to the concept of temperature in advanced CGT. Chapter 22 closes the report by implementing these concepts in C++.

Chapter 16

Graph Games

16.1 Quick Vocabulary

A graph is a set of dots (vertices) connected by lines (edges). If the edges have arrows, the graph is directed; otherwise it is undirected. A vertex's neighbours are whatever it connects to directly. A dead end is a vertex with no legal outgoing move — whoever is forced to move there loses under normal play. This handful of terms is genuinely all the vocabulary needed for the rest of the chapter.

16.2 Geography

> **Rules of Geography.** A token sits on a vertex. Each turn, the player to move slides the token along an edge to an adjacent vertex that has not been visited before. Once visited, a vertex is removed entirely from the graph, along with all of its edges. A player who cannot make a legal move loses.

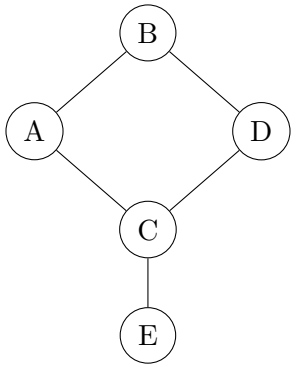
The name comes from the game Atlas: one player names a city, the next player's city must start with the previous city's last letter, and no repeats are allowed. Here the cities become vertices.

16.3 Why It Is the Centre of Graph Games

A Geography game is played on a graph $G = (V, E)$, where each vertex represents a location and each edge represents a legal move between two locations. Unlike Nim, where the game consists of independent heaps, Geography introduces the concept of movement constrained by graph topology. The structure of the graph determines which moves are available, making graph connectivity the central strategic element of the game.

16.4 Worked Example: Geography

Consider the following undirected graph. The token is initially placed at vertex A. Once a vertex is visited, it cannot be visited again. The player who is unable to make a legal move loses.



Assume Player 1 starts the game.

Move 1. Player 1 moves the token from A to B. Since A has now been visited, it is no longer available for future play.

Move 2. Player 2 moves from B to D. Returning to A is not permitted, because previously visited vertices cannot be revisited.

Move 3. Player 1 moves from D to C, leaving only one unexplored continuation.

Move 4. Player 2 moves from C to E. Vertex E has no unvisited neighbours, so the game reaches a terminal position.

Result. It is now Player 1's turn, but no legal move exists. Therefore, Player 2 wins the game.

16.5 Discussion and Conclusion

This example illustrates that the outcome of Geography is determined by the structure of the graph rather than simply by the number of remaining vertices. Every move permanently removes part of the graph from future consideration, changing the set of legal moves available in later stages of the game. The objective is therefore to guide the token into a position from which the opponent has no valid continuation.

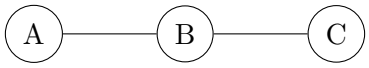
This simple example highlights why Geography is regarded as one of the fundamental graph games in combinatorial game theory. Unlike Nim, where the focus is on manipulating heap sizes, Geography requires players to reason about the changing connectivity of a graph. Each move reshapes the game by removing future possibilities, making long-term planning more important than immediate gain. This shift in perspective motivates the recursive techniques developed in the next chapter, where game positions are analysed using the Sprague–Grundy theorem.

Chapter 17

Applying Sprague–Grundy Theory to Graph Games

17.1 Worked Example: Grundy Value of a Path on Three Vertices

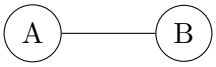
Consider the path graph P_3 shown below. A legal move consists of removing a vertex and all edges incident on that vertex.



Step 1 — Base Case. The empty graph contains no legal moves. Therefore its Grundy value is defined as $G(\emptyset) = 0$. This serves as the base case for the recursive computation.

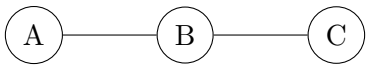
Step 2 — Single Vertex. Consider a graph containing only one isolated vertex. Removing that vertex leaves the empty graph, whose Grundy value is 0. The successor set is $\{0\}$; applying mex gives $\text{mex}\{0\} = 1$. Hence every isolated vertex has Grundy value $G(P_1) = 1$.

Step 3 — Two Connected Vertices.



Two legal moves are possible. Removing A leaves the isolated vertex B (Grundy value 1), while removing B leaves the isolated vertex A (also Grundy value 1). The successor Grundy values are $\{1, 1\}$. Applying mex gives $\text{mex}\{1\} = 0$. Therefore $G(P_2) = 0$.

Step 4 — Three Connected Vertices.



Three moves are possible:

- Remove A — the remaining graph is B—C, whose Grundy value is $G(P_2) = 0$.
- Remove C — by symmetry, the remaining graph again has Grundy value 0.
- Remove B — the graph splits into two independent isolated vertices, A and C. Each has Grundy value 1, and since the components are independent, their combined value is $1 \oplus 1 = 0$.

Thus, all three successor positions have Grundy value 0.

Step 5 — Applying mex. The successor Grundy values are $\{0, 0, 0\}$. Applying mex yields $\text{mex}\{0\} = 1$. Hence the Grundy value of the path on three vertices is $G(P_3) = 1$.

17.2 Discussion

This shows the practical use of Sprague–Grundy theory in graph games. Every legal move produces a new graph, and the Grundy value of the original graph is obtained by recursively evaluating these successor graphs. Whenever a move disconnects the graph into independent components, the Grundy values of those components are combined using XOR before applying mex.

17.3 Conclusion

This allows complex graph games to be analysed using the same mathematical framework originally developed for Nim. Although graph games are more complex than Nim, the recursive strategy remains unchanged. By evaluating successor graph positions, every position can be assigned an equivalent Nim value, making it possible to identify winning and losing positions through Sprague–Grundy theory rather than intuition alone.

This example proves that Sprague–Grundy theory helps analyze a wide variety of impartial games, including those played on arbitrary graph structures. This idea serves as the foundation for the graph-based applications discussed in the remainder of this report.

Chapter 18

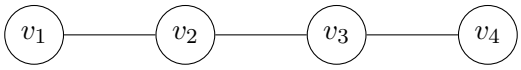
Node Kayles and Treblecross

18.1 Node Kayles

> **Rules of Node Kayles.** Pick any live vertex; remove it and every one of its neighbours from the graph. A player unable to move loses.

This differs from Geography in an important structural way: a single move can eliminate a whole cluster of vertices at once, and it typically splits the remaining graph into several disconnected pieces — which is exactly when the XOR rule from Chapter 3 becomes essential rather than optional.

Worked example: a path of four vertices, v_1–v_2–v_3–v_4.



- Picking v_1 removes v_1 and v_2, leaving a path of length 2 (v_3–v_4).
- Picking v_2 removes v_1, v_2, and v_3, leaving only v_4.
- Picking v_3 is symmetric to picking v_2.
- Picking v_4 is symmetric to picking v_1.

Smaller values are needed first. An empty path has $G = 0$. A path of one vertex: the only move empties it, so $G = \text{mex}\{0\} = 1$. A path of two vertices: either endpoint move empties it, so $G = \text{mex}\{0\} = 1$ as well.

So for the path of four vertices, all four possible moves land on a resulting Grundy value of 1 — as a set, just $\{1\}$. This gives:

$$G(P_4) = \text{mex}\{1\} = 0$$

The player forced to move first on a path of four vertices actually loses under optimal play from both sides — a fact that is not obvious just from looking at the board, and a good illustration of why Grundy analysis is necessary rather than just convenient.

18.2 Treblecross

18.2.1 Rules

> **Rules of Treblecross.** A row of n boxes, each either empty or marked X. A move marks one empty box. Completing three consecutive X's wins the game outright.

18.2.2 Splitting Into Runs

Once a few cells are marked, the empty boxes break into separate stretches ("runs") that no longer interact with each other, and — as in every game so far — each run can be solved on its own and the results combined with XOR. So in principle there is just one function to determine: $G(n)$, the Grundy value of an isolated run of n empty boxes.

18.2.3 What Makes Treblecross Special

While studying the recursive computation of Grundy numbers for Treblecross, my first intuition was that the Grundy value of a position should depend only on the length of the available empty interval — that two intervals of the same length would always have the same Grundy value, irrespective of what lay outside the interval. However, after working through a few examples, I realised this assumption is not correct.

To understand why, consider the following position:

```
X . X
```

The middle cell is empty, while both neighbouring cells already contain an X. At first glance, this appears to be just a single empty square. However, if a player places an X in the middle cell, the board immediately becomes

```
X X X
```

which forms three consecutive marks and ends the game instantly. This example shows that the legality and strategic value of a move cannot be determined by looking only at the empty interval — the neighbouring cells outside the interval also play an important role.

A similar situation arises when an empty interval is adjacent to a single existing X. Consider the position

```
X . .   .
```

There is no existing XX pair, so the interval may initially appear safe. However, if a player marks the first empty cell, the position changes to

```
X X . .
```

Although the game has not ended yet, the newly created XX pair immediately makes the next cell dangerous, because playing there would complete XXX. Thus, the presence of even a single external X changes the future evolution of the game by restricting the set of safe moves.

These examples demonstrate that two empty intervals of the same length are not necessarily equivalent. Their behaviour depends not only on the number of empty cells but also on the condition of their boundaries. An interval surrounded by empty cells behaves differently from one that is adjacent to existing marks.

Therefore, while computing Grundy numbers recursively, it is not sufficient to describe a game state solely by the length of the interval — the state of the left and right boundaries must also be considered. Ignoring these boundary conditions would incorrectly treat different game positions as identical, leading to incorrect Grundy values and an inaccurate recursive analysis.

18.2.4 Conclusion

In Nim, a move is always legal, and a pile's value depends only on its size — nothing about its history matters. In Node Kayles, a move is always legal too; the graph simply gets smaller. Treblecross is the first game here where a move can be illegal depending on its boundary cells.

That single fact is what pushes Treblecross out of the "obviously equals a Nim heap" category and into the family known as octal games: row games where the moves and their consequences

are governed by local patterns of marks. Octal games each get a short code describing exactly what a move is allowed to do; Treblecross's code is 0.007.

Most small octal games, once their Grundy sequence is computed far enough, settle into a pattern that repeats forever, and this eventual periodicity is provable for many of Treblecross's close relatives. Treblecross itself has resisted this completely. Researchers have computed its Grundy sequence out to boards with millions of cells, and as of the most recent published work on the subject, it remains an open problem whether the sequence is ultimately periodic at all — a genuinely striking fact for a game whose entire rule set is "place an X, don't make three in a row." This is precisely why Treblecross is singled out and studied on its own, rather than being treated as just another entry in a long list of octal games.

Because a run's value depends on both its length and the marked/open status of each boundary, the fully correct recurrence needs to track that boundary information, not just a single function $G(n)$.

Chapter 19

Toads and Frogs

19.1 Rules

Rules of Toads and Frogs. A strip of n boxes; some contain a Toad (T), some a Frog (F), the rest are empty. Toads belong to Left and only move rightward; Frogs belong to Right and only move leftward. On a turn, a player slides their piece into an adjacent empty box, or jumps it over exactly one enemy piece into the empty box directly behind it (the jumped piece is unaffected and stays where it is). A player with no legal move loses.

19.2 A Full Worked Example: T.TF

19.2.1 Level 0 — the two terminal positions

- `F.TT` (Frog, gap, Toad, Toad) — both toads are blocked: the toad at box 3 faces an ally at box 4, and the toad at box 4 is at the right edge with nothing to move into. Right's frog is at the left edge and cannot move left. Neither side has a move, so `F.TT` $= 0$.
- `.TTF` (gap, Toad, Toad, Frog) — neither toad can advance (one is blocked by an ally, the other faces the frog with no jump landing available), and the frog cannot slide (blocked) or jump (no empty landing square beyond the blocking toad). Neither side has a move, so `.TTF` $= 0$.

19.2.2 Level 1

`.FTT` (empty, Frog, Toad, Toad) — Left's toads are still both stuck for the same reasons as above, so Left has no move. Right's frog can slide left into the empty box 1, reaching `F.TT` (value 0). So:

$$\texttt{.FTT} = \{\mid 0\} = -1 \quad \text{(Right wins irrespective of who moves first)}$$

19.2.3 Level 2 — TF.T and TFT.

Here is the full chain, worked outward from the position with the frog sandwiched between the two toads:

`TF.T` (Toad, Frog, gap, Toad) — Left's toad at box 1 jumps over the frog at box 2, landing in the empty box 3 (the frog stays put), reaching `.FTT`, value -1. Left's toad at box 4 is at the right edge and cannot move. Right's frog at box 2 cannot slide (box 1 is occupied) and cannot jump (no landing square past the left edge), so Right has no move. This gives:

$$\texttt{TF.T} = \{-1 \mid\} = 0 \quad \text{(whoever moves first loses)}$$

`TFT.` (Toad, Frog, Toad, gap) — Left's toad at box 1 cannot slide (blocked by the frog) or jump (landing square, box 3, is occupied by the other toad). Left's toad at box 3 slides into the empty box 4, reaching `TF.T`, value 0. Right's frog at box 2 cannot slide (box 1 occupied) and cannot jump (landing square, box 3, is occupied), so Right has no move. This gives:

$$\texttt{TFT.} = \{0 \mid\} = 1 \quad \text{(Left wins irrespective of the turn)}$$

19.2.4 Level 3 — the original position

`T.TF` (Toad, gap, Toad, Frog) — Left's toad at box 1 slides into the empty box 2, reaching `.TTF`, value 0. Left's toad at box 3 cannot move (blocked by the frog, and there is no landing square beyond the right edge for a jump). Right's frog at box 4 jumps over the toad at box 3, landing in the now-empty box 2, reaching `TFT.`, value 1. So:

$$\texttt{T.TF} = \{0 \mid 1\}$$

Interpreting the Result. As this position doesn't evaluate to 0, 1, or -1, it cannot simply be interpreted with our current knowledge — which leads directly to the concept of infinitesimals and tiny–miny numbers.

19.3 Where Grundy Numbers Actually Fail

Take `TF` again, and imagine (incorrectly) treating it as impartial, where either player could move either kind of piece. Mex would happily produce a single number for it. That number would be a lie, because who is actually favoured depends on which pieces belong to which player, not merely on the count of available moves. Some Toads-and-Frogs positions are "confused with zero" — neither side has a built-in edge, and the outcome depends entirely on who moves first. No single Grundy number can express that; only the two-sided $\{G^L \mid G^R\}$ structure can.

19.4 Numerical Game Values

19.4.1 Motivation

We were unable to analyse the result $G = \{0 \mid 1\}$ in the Toads and Frogs game, which motivated the study of numerical game values, infinitesimal games, and tiny–miny games.

19.4.2 Conway Giving Numerical Values to Games

One of the most important ideas introduced by John Conway is that certain games can be treated just like ordinary numbers. The motivation is simple: while studying different games, it is not enough to know whether a position is winning or losing — two winning positions may not be equally good, since one may give Left a much larger advantage than the other. Conway therefore introduced the concept of game numbers to measure how much a game favours a player.

Consider a Conway game $G = \{L \mid R\}$, where L is the set of positions Left can move to and R is the set of positions Right can move to. A game is assigned a numerical value only if there exists a real number x that lies strictly between every Left option and every Right option:

$$L < x < R$$

If such a number exists, the game behaves exactly like the real number x, and we assign that value to the game.

For example, $\{0 \mid 1\}$ is assigned the value $\frac{1}{2}$, because $0 < \frac{1}{2} < 1$. Similarly, $\{1 \mid 3\}$ is assigned the value 2, since $1 < 2 < 3$.

19.4.3 Examples of Numerical Games

- $1 = \{0 \mid\}$ — Left has a move to the neutral game 0, while Right has no legal move at all; this represents one full unit of advantage for Left.
- $2 = \{1 \mid\}$ — represents two full units of advantage for Left.
- $-1 = \{\mid 0\}$ — represents one full unit of advantage for Right, by symmetry.
- $\frac{1}{2} = \{0 \mid 1\}$ — Left can move to 0 and Right can move to 1, so the value lies exactly halfway between them.

These examples confirm that integers and fractions are simply special cases of Conway's general game notation, and that numerical values quantify how much one player is ahead under optimal play.

19.4.4 Summary

Numerical games behave exactly like ordinary real numbers under comparison and addition. However, as the next section shows, not every game produced by Conway's construction is numerical.

19.5 Beyond Numerical Values

19.5.1 Motivation

Although many games correspond to real numbers, this is not true in general. Some game trees produce positions whose outcome depends entirely on who moves first rather than on any measurable numerical advantage. Two such positions, star and Up, illustrate this limitation.

19.5.2 The Game Star

$$* = \{0 \mid 0\}$$

If Left moves first, Left wins by moving to 0, since Right is then left without a legal move. If Right moves first, Right wins for the same reason. Because the outcome depends entirely on the order of play rather than on a fixed numerical advantage, no real number can accurately represent this game — there is no real number that is simultaneously more than zero and less than zero.

19.5.3 The Game Up

$$\uparrow = \{0 \mid *\}$$

This game is always favourable to Left, yet, like star, it cannot be assigned any real numerical value, because we would need an x strictly less than $*$, and since $*$ has no real value, no such x can be found. Its presence in Conway's framework motivates the introduction of a new class of values altogether.

19.5.4 Summary

Games such as star and Up demonstrate that Conway's recursive notation generates positions that lie beyond the ordinary real number system. This limitation motivates the infinitesimal game values developed in the remainder of this report.

19.6 Infinitesimal Game Values

19.6.1 Definition

An infinitesimal game is a position whose value is greater than zero but smaller than every positive real number. Formally, a game G is infinitesimal if

$$-\varepsilon < G < \varepsilon \quad \text{for every positive real number } \varepsilon$$

19.6.2 Interpretation

Unlike the game 0, an infinitesimal is not a neutral position. It represents a very small strategic advantage arising purely from initiative. Rather than measuring how much a player is ahead, infinitesimals measure who benefits from moving first.

19.6.3 Importance

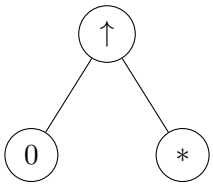
Infinitesimal values extend Conway's theory well beyond the ordinary number line and are essential for describing endgame positions in games such as Domineering and Hackenbush, where small residual advantages frequently arise from initiative alone.

19.7 The Up Game (↑)

19.7.1 Definition and Game Tree

The simplest positive infinitesimal is the Up game:

$$\uparrow = \{0 \mid *\}$$



19.7.2 Why Up Favours Left

If Left moves first, Left immediately moves to 0, ending the game favourably. If Right moves first, Right can only move to $*$. Since $*$ is a first-player winning game, Left now becomes the first player within $*$ and eventually wins. Thus the Up game always favours Left, regardless of who moves first.

19.7.3 Why Up Is Not Equal to 1

Although Up always favours Left, it is not equivalent to the numerical value 1. Recall that $1 = \{0 \mid\}$ represents a complete unit of advantage, because Right has no legal move at all, so Left's advantage cannot be challenged in any way.

In contrast, in $\uparrow = \{0 \mid *\}$, Right does have a legal move. Although this move cannot change the eventual winner, it gives Right an opportunity to influence the game before losing. Up therefore represents initiative rather than a complete numerical advantage, and Left's advantage is strictly weaker than a full unit:

$$0 < \uparrow < 1$$

19.7.4 Up as the Canonical Infinitesimal

In fact, a stronger statement holds: $0 <\uparrow< \varepsilon$ for every positive real number ε. This makes Up the canonical example of a positive infinitesimal, and it serves as the reference value against which other infinitesimal games are measured.

19.8 Down, Tiny, and Miny Games

19.8.1 The Down Game

The mirror image of Up is the Down game:

$$\downarrow = \{* \mid 0\}, \qquad \text{satisfying } \downarrow = -\uparrow$$

Down provides an infinitesimal advantage to Right in exactly the same way that Up provides one to Left.

19.8.2 Tiny Games

After introducing the Up game, Conway extended the idea of infinitesimal values by defining a broader family known as Tiny games, commonly represented in the simplified form

$$\{0 \mid -x\}, \quad x > 0$$

Tiny games represent positive infinitesimal values, meaning they provide a very small strategic advantage to Left. Unlike ordinary numerical games, this advantage cannot be measured by any positive real number — instead it reflects a subtle positional edge arising from the structure of the available moves and the initiative possessed by the players.

Tiny games therefore generalize the concept of the Up game and demonstrate that positive infinitesimal values are not limited to a single example.

19.8.3 Miny Games

The corresponding family of Miny games forms the negative counterpart of Tiny games, commonly represented as

$$\{-x \mid 0\}, \quad x > 0$$

Miny games represent negative infinitesimal values, providing a small strategic advantage to Right. Just as Tiny extends the family of positive infinitesimals, Miny extends the family of negative infinitesimals.

Together, Tiny and Miny show that infinitesimal values form an infinite hierarchy of game values rather than being restricted to the simple Up and Down games.

19.8.4 Importance of Tiny and Miny

The introduction of Tiny and Miny significantly expands Conway's theory of infinitesimal games. While Up and Down provide the simplest examples of positive and negative infinitesimals, Tiny and Miny demonstrate that there exist infinitely many such values with increasingly subtle strategic behaviour.

These games highlight an important limitation of ordinary real numbers: not every strategic advantage in a combinatorial game can be represented by a measurable numerical quantity. Instead, many positions derive their significance from initiative and move order, requiring richer mathematical objects than ordinary numbers. Consequently, Tiny and Miny play an important role in extending Conway's framework beyond numerical games and illustrate the depth and richness of partizan game theory.

19.9 Conclusion

The study began with Toads and Frogs, where Sprague–Grundy theory failed because the game was no longer impartial, so Conway's theory was used to analyse it instead. The Conway notation $G = \{0 \mid 1\}$ could not simply be interpreted for who was winning, which led to how Conway represented game values as familiar integers and fractions, confirming that numerical values are simply a subset of all possible games. It then explored the limitations of numerical values through games such as $*$ and $\uparrow$, introducing infinitesimal values that capture initiative rather than measurable advantage. Finally, the Up, Down, Tiny, and Miny games illustrated that many strategic positions cannot be adequately described using ordinary numbers alone.

These ideas provide the theoretical foundation for temperature theory, which is taken up in a later chapter and extends this analysis to the broader question of how urgently a move should be played.

Chapter 20

Blue–Red Hackenbush

20.1 Rules, Gravity Included

> **Rules of Blue–Red Hackenbush.** Colored line segments (blue or red) stand on a horizontal ground line, sometimes stacked on or branching from each other. Left may delete any blue segment; Right may delete any red segment. After a deletion, any segment no longer connected to the ground, even indirectly, falls away and disappears too (the gravity rule). A player with no legal move loses.

Gravity is the whole trick that makes Hackenbush interesting: a single deletion can wipe out an entire structure above it, which has no analogue in Nim, where removing tokens from one pile never touches anything else.

20.2 Examples

A single blue segment, alone, standing on the ground — Left deletes it, leaving nothing; Right has nothing to touch. So $G = \{0 \mid\}$: Right literally never gets to move here, a guaranteed Left win.

A blue segment stacked on a red one — Left can only take the top (blue) piece, leaving a lone red standing. Right can only take the bottom (red) piece, but gravity drops the blue one along with it, so Right's single move is worth removing two segments at once. This is the cleanest illustration of why gravity matters.

One blue and one red segment, standing apart (not stacked) — this is just the sum of the two single-segment cases above. Left takes blue, leaving red; Right takes red, leaving blue. The first mover wins, so this position is $*$, though here it arises from an actual colour mismatch rather than a repeated option.

20.3 How a Stack Actually Turns Into a Number

Start with the two simplest stacks: a lone blue edge on the ground, worth $G = \{0 \mid\}$ as shown above. Left can always move (delete it); Right can never move at all. That is about as one-sided as a game gets, and it lines up exactly with the number $+1$ — Left is permanently one move ahead, with no way for Right to ever catch up. By the mirror argument, a lone red edge is worth -1.

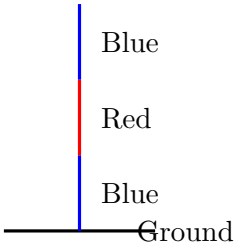
If n blue edges are stacked directly on top of each other, nothing changes qualitatively: Left simply deletes from the top down, one per turn, and Right never gets a look in. That stack is worth $+n$. A stack of n reds, by the same logic, is worth $-n$. So a single-colour stack is just an integer equal to its height, positive for blue and negative for red — nothing mysterious yet.

The interesting behaviour starts the moment a stack changes colour partway up. Take a two-edge stack, red on the bottom, blue on top. Its value is $\{-1 \mid 0\}$ (Left's only move leaves a lone red, worth -1; Right's only move drops everything, worth 0). The Simplicity Rule says the value must be the simplest number strictly between -1 and 0, which is $-1/2$. So a red-then-blue stack is worth exactly $-1/2$.

Flipping the colours — blue on the bottom, red on top — gives, by the mirror argument, $G = \{0 \mid 1\}$, and the simplest number strictly between 0 and 1 is $1/2$. So a blue-then-red stack is worth $+1/2$.

20.4 A Worked Example: Computing the Value of a Blue–Red–Blue Stack

To understand how numerical values arise in Blue–Red Hackenbush, consider the following simple graph.



We now compute the value of this position using Conway's recursive game notation.

Suppose Left moves first. Since Left controls only the blue edges, two possible moves exist.

15. Removing the bottom blue edge disconnects the entire graph from the ground, causing every remaining edge to fall. The resulting position is the empty game, with value 0.

16. Removing the top blue edge leaves only the lower blue and red edges — a blue-then-red stack, which was shown in Section 8.3 to have value $+1/2$.

Hence, the set of positions available to the Left player is $\{0, 1/2\}$.

Now consider Right. Since Right controls only the red edge, there is only one possible move: removing the red edge disconnects the top blue edge (which falls away), leaving only the bottom blue edge attached to the ground — a single blue edge, of value 1.

Therefore, Right's set of options is $\{1\}$. Combining these results, the game can be written as:

$$G = \{0, \tfrac{1}{2} \mid 1\}$$

According to Conway's Simplicity Rule, the value of a game is the simplest number lying strictly between the largest Left option and the smallest Right option. Since the largest Left option is $1/2$ and the smallest Right option is 1, the simplest number strictly between them is $3/4$. Thus, the value of the Blue–Red–Blue stack is:

$$G = \tfrac{3}{4}$$

An Interesting Pattern. After computing several similar examples, an interesting pattern begins to emerge. Instead of carrying out the complete recursive analysis every time, the value of a vertical Hackenbush stack can often be determined directly from its structure.

The procedure is remarkably simple. Starting from the ground, first count the consecutive edges of the same colour — these edges contribute the whole-number part of the game value, blue contributing positively and red contributing negatively. Whenever the colour changes, the

contribution of each subsequent edge is halved: the first edge after the colour change contributes $\pm 1/2$, the next contributes $\pm 1/4$, then $\pm 1/8$, and so on, with the sign determined by the colour of the corresponding edge.

Applying this rule to the previous example (blue, red, blue from the ground up):

- Bottom blue edge: $+1$
- Middle red edge: $-1/2$
- Top blue edge: $+1/4$

Therefore the total is $1 - 1/2 + 1/4 = 3/4$, exactly the same value obtained using Conway's recursive definition.

This shortcut makes it possible to evaluate simple Hackenbush stacks much more efficiently while still producing the same result as the full recursive analysis. It also reveals an important property of Blue–Red Hackenbush: every finite vertical stack corresponds to a dyadic rational number — a fraction whose denominator is a power of two. However, this shortcut applies only to simple stacks. A question still remains: how urgently does that move need to be made, and can delaying it change the game? This leads to the concept of temperature in CGT.

Chapter 21

Concept of Temperature in CGT

21.1 Introduction

The primary objective of this chapter is to study temperature theory, one of the most important concepts in combinatorial game theory. It introduces the concepts of hot and cold games, mean value, game temperature, move urgency, cooling, and heating, which together provide a mathematical framework for analysing both the value of a game and the urgency of making a move.

21.2 Hot and Cold Games

Before introducing temperature theory, it is essential to distinguish between hot and cold games. These concepts describe whether the timing of a move affects the strategic value of a game.

A cold game is one in which delaying a move has little or no strategic consequence — the value of the position remains essentially unchanged regardless of who moves first. Such games primarily represent stable numerical advantages rather than initiative.

For example, the numerical game $1 = \{0 \mid\}$ always provides Left with one complete unit of advantage. Whether Left moves immediately or waits, the underlying value of the position remains the same. Similar behaviour is observed for other numerical games such as 2, $\frac{1}{2}$, and -3. These are therefore classified as cold games.

In contrast, a hot game is one in which moving first provides an immediate strategic benefit — delaying a move may result in losing an important tactical opportunity, making the order of play highly significant.

Examples include $* = \{0 \mid 0\}$ and $\uparrow = \{0 \mid *\}$. In the game $*$, the first player always wins, making initiative the deciding factor. Similarly, the Up game provides Left with a small advantage because moving first preserves initiative.

Thus, hot games emphasize initiative, whereas cold games emphasize positional value.

Although this classification indicates whether a game is strategically urgent, it does not explain how urgent a move actually is. Two different games may both be hot, yet one may require immediate attention while another can safely be postponed. This limitation motivates the development of temperature theory.

21.3 Introduction to Temperature Theory

Consider a position consisting of several independent games, all favourable to the current player. The natural question is: which game should be played first?

Merely knowing that a game is hot does not answer this question. Some games lose a significant amount of value if ignored, while others lose very little.

To solve this problem, John Conway introduced temperature theory, which assigns a numerical measure to the urgency of making a move. Temperature theory separates every game into two independent components:

- **Mean value** — the long-term positional worth of the game.
- **Temperature** — the urgency associated with making the next move.

Together, these two quantities provide a much richer description of a game than its numerical value alone.

21.4 Mean Value

The mean value represents the average positional strength of a game after ignoring the urgency associated with moving first. For a simple switch game $G = \{L \mid R\}$, where L and R represent the best options available to Left and Right respectively, the mean value is:

$$m = \frac{L + R}{2}$$

The mean may be interpreted as the stable numerical value around which the game fluctuates. For example, $G = \{4 \mid 2\}$ has the mean $m = (4 + 2)/2 = 3$. This indicates that the position is approximately worth three units irrespective of whose turn it is — the mean describes the long-term positional advantage of the game.

21.5 Game Temperature

While the mean describes the overall value of a game, it does not measure the importance of making the next move — this role is performed by the temperature. For a simple switch game $G = \{L \mid R\}$, the temperature is defined as:

$$t = \frac{L - R}{2}$$

Temperature measures the difference between the best options available to the two players. Returning to the previous example, $G = \{4 \mid 2\}$ has temperature $t = (4-2)/2 = 1$. This means that although the position is worth approximately three units, delaying a move may result in losing roughly one unit of strategic value.

Games with high temperature are highly tactical and require immediate attention, whereas games with low temperature can safely be postponed.

21.6 Move Urgency

One of the most significant consequences of temperature theory is the concept of move urgency. Suppose a player has several independent games available during a turn. Although each game may be favourable, they need not be equally urgent.

Consider two independent games:

- Game A — temperature = 6
- Game B — temperature = 1

Even if both games have similar positional values, Game A should always be played first, because delaying it causes a much larger loss of value.

This principle forms the basis of optimal play in many advanced combinatorial games. Instead of choosing moves solely based on positional value, players prioritize the hottest remaining game, thereby preserving the greatest possible strategic advantage. Temperature therefore provides a quantitative method for deciding the order in which different games should be played.

21.7 Cooling a Game

To better understand game temperature, Conway introduced the concept of cooling — the process of gradually reducing the urgency associated with making a move while preserving the underlying positional value of the game.

Conceptually, cooling may be viewed as introducing a small "tax" on every move. As this tax increases, both players obtain progressively less benefit from making an immediate move.

For a simple switch game $G = \{L \mid R\}$, cooling the game by an amount c produces:

$$G_c = \{L - c \mid R + c\}$$

The difference between the players' options gradually decreases, causing the temperature of the game to fall. Eventually, the temperature becomes zero — at this stage, the game behaves exactly like an ordinary numerical game, retaining only its mean value.

Cooling therefore separates initiative from positional advantage, making it one of Conway's most elegant analytical tools.

21.8 Heating a Game

Heating is the inverse operation of cooling. Instead of reducing urgency, heating increases the strategic importance of making a move — the difference between Left's and Right's available options becomes larger, increasing the temperature of the game.

Heating allows a previously cooled game to regain its tactical significance, and is useful when reconstructing the original game from its cooled form.

Together, cooling and heating demonstrate that every hot game can be viewed as a combination of a stable positional value and a varying degree of initiative.

Chapter 22

Computational Aspects of Combinatorial Game Theory using C++

22.1 Introduction

In this final chapter, the focus shifts from the theoretical concepts of combinatorial game theory developed throughout this report to their computational implementation using C++, translating impartial games, the Sprague–Grundy theorem, and Nim into simple programs that can analyze game positions and determine winning strategies.

22.2 The mex Function

The first task involved implementing the Minimum Excluded Value (mex) function. The mex of a set is defined as the smallest non-negative integer that is not present in that set. Since the computation of Grundy numbers is based on the mex operation, it forms one of the most fundamental components of impartial game algorithms.

```cpp
#include <iostream>
using namespace std;

int mex(int arr[], int n)
{
    int m = 0;
    while (true) {
        int found = 0;
        for (int i = 0; i < n; i++) {
            if (arr[i] == m) { found = 1; break; }
        }
        if (found == 0) return m;
        m++;
    }
}

int main()
{
    int arr[] = {0, 1, 3};
    cout << "mex = " << mex(arr, 3);
    return 0;
}
```

22.3 Computing Grundy Numbers

After implementing the mex function, it was used to compute Grundy numbers for a simple subtraction game in which a player is allowed to remove either one or two stones during each turn. For every game position, the Grundy number is obtained by taking the mex of the Grundy numbers of all positions that can be reached in one move.

```cpp
#include <iostream>
using namespace std;

int mex(int arr[], int n)
{
    int m = 0;
    while (true) {
        int found = 0;
        for (int i = 0; i < n; i++) {
            if (arr[i] == m) { found = 1; break; }
        }
        if (found == 0) return m;
        m++;
    }
}

int grundy(int n)
{
    if (n == 0) return 0;
    int g[2];
    g[0] = grundy(n - 1);
    if (n >= 2) g[1] = grundy(n - 2);
    else g[1] = -1;
    return mex(g, 2);
}

int main()
{
    for (int i = 0; i <= 10; i++)
        cout << "Grundy(" << i << ") = " << grundy(i) << endl;
    return 0;
}
```

The generated Grundy numbers classify each game position as either winning or losing. Positions with Grundy number 0 are losing positions under optimal play, whereas positions with non-zero Grundy numbers are winning positions. This implementation demonstrates how the mex function can be directly used to compute Grundy numbers recursively for a simple impartial game.

22.4 Application to Nim

The concepts of mex and Grundy numbers were then applied to Nim, one of the most important impartial combinatorial games. In Nim, each heap has a Grundy number equal to its size, and the overall game is analyzed by computing the nim-sum, the bitwise XOR of the sizes of all heaps. If the nim-sum is zero, the current player is in a losing position; otherwise, the current player has a winning strategy.

```cpp
#include <iostream>
using namespace std;

```

```cpp
int main()
{
    int heap[] = {3, 4, 5};
    int nimSum = 0;
    for (int i = 0; i < 3; i++)
        nimSum = nimSum ^ heap[i];
    if (nimSum == 0)
        cout << "Second Player Wins";
    else
        cout << "First Player Wins";
    return 0;
}
```

Note: the nim-sum is the bitwise XOR (^) of the heap sizes, not their arithmetic sum — the operator has been corrected here accordingly.

22.5 Conclusion

Implementing these algorithms provided practical insight into how the concepts studied in previous chapters can be translated into computer programs. The mex function served as the building block for Grundy number computation, while the Nim implementation demonstrated one of the simplest and most elegant applications of the Sprague–Grundy theorem.

Closing Remarks

Beginning with P-positions and Nim and closing with a working C++ implementation of the Sprague–Grundy algorithm, this report has traced a single continuous thread: recursive analysis turns an enormous variety of games — piles, graphs, rows of marked cells, colored line segments, and strips of Toads and Frogs — into a small number of well-understood mathematical objects: Grundy numbers, nimbers, surreal numbers, and temperatures. Future study can extend this thread further, into infinite games, wider classes of octal games, and increasingly general computational tools.

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